Fiscal Year:	FY 2015	**Task Last Updated:**	FY 06/12/2015
PI Name:	Kucik, Dennis F. M.D., Ph.D.		
Project Title:	Radiation-Induced Early Changes in Gene and Protein Expression, Lipid Oxidative States, and Vascular Function Related to Atherosclerosis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35205-4831	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	08/01/2014	**End Date:**	07/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kabarowski, Janusz (University of Alabama, Birmingham) White, Charles (University of Alabama, Birmingham)		
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Task Description:	Epidemiological studies have established that radiation from a number of terrestrial sources increases the risk of atherosclerosis (a cause of heart attack and stroke), even at relatively low doses. The effect of galacto-cosmic radiation (GCR), such as encountered on deep space missions, is not clear, however, because this kind of radiation interacts with tissues very differently than does most terrestrial radiation. In our previous project, we showed that heavy ions, an important component of GCR, indeed accelerate the development of atherosclerosis-related plaques in the mouse, but only at relatively high doses. Mouse models, however, are known (from comparison of animal experiments to human epidemiological data) to require much higher radiation doses for plaque formation than humans, so while this type of experiment yields valuable information about the effect of radiation, it is not very useful in predicting risk at a given dose. This is likely due to species differences in lipid metabolism and the immune system. Plaque formation is a very late endpoint, however. Very early steps in atherosclerosis, such as changes in gene expression, protein expression, and lipid oxidation, should be much more similar in humans and mice. Analysis of these early changes in response to doses of simulated GCR that astronauts are likely to encounter on a Mars mission will enable us to better predict the risk of cardiovascular consequences from space radiation while it increases our understanding of the mechanisms involved. Our hypothesis is that irradiation of blood vessels leads to early, species-independent changes in endothelial cell genes, proteins, and lipids that cause vascular dysfunction and are therefore likely to be predictive of cardiovascular disease. An understanding of these changes can be used to help predict risk and propose countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Atherosclerosis is a major adverse effect of therapeutic radiation. It is not always possible to avoid exposure of the heart and major arteries, especially for treatment of head-and-neck cancers and breast cancer. This project will provide important new information about the mechanism of radiation-induced atherosclerosis to help in designing possible mitigating strategies. Moreover, the use of ion beams for cancer treatment is becoming ever more prevalent. In the US, proton therapy is becoming more widespread, and carbon-ion therapy is available in Europe and Asia. Little is known about the cardiovascular effects of these new radiation treatment modalities. This work will greatly enhance our understanding of possible adverse effects.
Task Progress:	As this was the first year of this project, limited results are available at this time. We have performed three irradiations at the NASA Space Radiation Laboratory (NSRL), but analysis of the irradiated tissues and fluids is still in early stages. We were, however, able to make two important early findings. First, we were able to demonstrate that accelerated iron ions, an important component of galacto-cosmic radiation, has early effects on adhesiveness of the inner surface (endothelium) of arteries. This is important because such changes in endothelial adhesiveness are an early event in the development of atherosclerosis. In fact, there is good evidence that prevention of these changes (in mice) is protective, decreasing the chance that atherosclerosis will ever progress to the point of mature plaques that can rupture and lead to myocardial infarction or stroke. Thus, a further understanding of this process may eventually lead to identification of countermeasures that could protect astronauts from the cardiovascular consequences of radiation exposure. Further, knowledge gained from this line of investigation might be useful in preventing or mitigating cardiovascular effects of therapeutic radiation on Earth, a major adverse effect of radiation treatments for cancer. Second, we were able to demonstrate by proteomic methods that radiation causes a persistent change in the levels of certain proteins in blood. These changes have important implications for the mechanism of radiation-induced atherosclerosis, as they mostly affect a particular component of the multi-factorial mechanism of the disease. While we have, so far, only analyzed samples from x-irradiated mice, we have collected samples from proton- and iron ion-irradiated animals as well. Therefore, we should soon be able to determine how these space-related forms of radiation affect the protein composition of circulating blood in ways that have implications for the risk of cardiovascular consequences of interplanetary travel.
Bibliography Type:	Description: (Last Updated: 04/12/2018)
Abstracts for Journals and Proceedings	Babitz SK, Dunaway CW, Steele C, Kucik DF. "Radiation results in IL-8 mediated intercellular signaling that increases Adhesion between monocytic cells and Aortic Endothelium." Presented at the Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. , Aug-2014
Abstracts for Journals and Proceedings	Babitz SK, Dunaway CW, Steele C, Kucik DF. "Involvement of IL-8 in radiation-induced, atherosclerosis-related adhesiveness changes of arterial endothelium." Presented at the 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. Abstract book. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. , Sep-2014
Abstracts for Journals and Proceedings	Babitz SK, Dunaway CW, Steele C, Kucik DF. "IL-8 mediated intercellular signaling following 56Fe irradiation results in increased aortic endothelial adhesiveness." Presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Chanda D, Gupta K, Kabarowski JH, Kucik DF. "56Fe irradiation of wild type C57BL/6 mice results in increased adhesiveness of aortic endothelium." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Abstracts for Journals and Proceedings	Kucik DF, Chanda D, Gupta K, Kabarowski JH. "Space-relevant dose of 56Fe radiation results in a pro-atherogenic change in the aortic endothelium of wild type C57BL/6 mice." Presented at the Space Radiation and Heavy Ions in Therapy Symposium 2015, Osaka, Japan, May 22-24, 2015. Space Radiation and Heavy Ions in Therapy Symposium 2015, Osaka, Japan, May 22-24, 2015. , May-2015

Articles in Peer-reviewed Journals	White RC, Yu T, Gupta K, Babitz SK, Black LL, Kabarowski JH, Kucik DF. "Early changes in vascular reactivity in response to 56Fe irradiation in ApoE -/- mice." Acta Astronautica. 2015 Mar-Apr;108:40-5. http://dx.doi.org/10.1016/j.actaastro.2014.10.010 , Apr-2015